SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1514

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I B.V. Tharun (192525155) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tharun B.V

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

1. Define clod computing and Explain ~~my~~ four Chatacteristics that Make it Different from Traditional on - premise computing:

Definition:- cloud computing mode in which resources such as servers, storage, databases, networking, software, and processing power are provided to users through the internet on demand. Istead of purchasing and maintaining physical infrastructe, organizations can obatin computing resource from cloud service providers accordingly according to their requirement.

FOur Chasracteristics of Cloud Computing:-

1. On - Bemand Self-Service:-

users can independently computing resources wherever they are required. for example a user can create a virtual server or increase storage through a cloud management console without waiting for manual installation.

2. Broad Network ACCESS:-

cloud services can be accessed through the internet using different devices such as Computing, laptops, tables, and smartphones. This allows users to access applications and data from different locations.

3. **Resource Pooling:-** Cloud providers maintain a shared pool of computing recouras such as CPU, memory, storage, and network bandwidth. These resources are dynamically allocated to multiple users according to their requirements.

4. **Rapid Elasticity:-** Cloud resources can be quickly increased or decreased according to workload for example, an online shopping website can add more computing resources during a festival sale and reduce them when demand decreass.

Cloud Computing	On-premise computing
* Resources are accessed through the internet	* Resources are maintained by the organization
* Lower intial infrastructure investment	* high initial infrastructure inverstment
* Resources can be scaled quickly	* Scaling requires additional hardcoare.

Conclusion:- Cloud computing provides flexibility, scalability, accessibility-, and efficient resource utilization. Its characteristic make it more adaptable than traditional-on-premise computing and suitable for modern applications and organizations.

2. Trace the Evolution of cloud computing from hardware Evolution to Internet Software Evolution and Evaluation and show how these changes Enabled modern cloud platforms.

Introduction :- To cloud computing is the result of several decades of technological development it evolved from centralized main frame computers to client-server system, internet technologies distributed computing, virtualization, and web-based software. each stage introduced technologies that contributed to the development of modern cloud platforms.

1. Main frame Computing :-

main frame were powerful centrazed computers used by multiple users through terminals they introduced the concept of centralized processing and shared computing resources.

2. Client-server computing :-

the client-server model divided computing tasks between client computers and servers Servers provided applications, processing and data to multiple clients through a network.

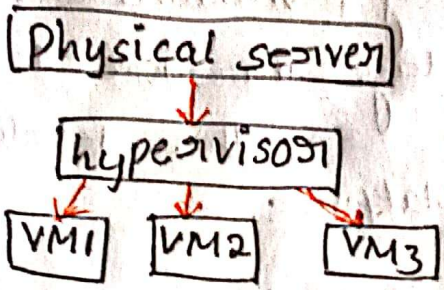
3. Internet Development :- The growth of the internet allowed computers located in different geographical locations to communicate.

4. Distributed Computing :-

Distributed computing allowed multiple computing to work together to perform tasks workloads could be distributed across system improving performance and reliability.

5. Virtulization :-

virtualization enabled one physical server to run multiple virtual machines.



Modern Cloud platforms :-

modern cloud computing combines Virtualization, high-speed internet, distributed Computing, automating, large-scale data centers and web technologies.

Conclusion :-

* Cloud computing evolved gradually from hardware and networking technologies to internet-based software services.

* Each stage improved resource sharing Connectivity, scalability and accessibility

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Introduction:- server virtualization is a fundam technology used in cloud computing. In traditional computing, one physical server many be dedicated to one application, which can result in underutilization, virtualization allow servel virtual servers to run to on a single physical server.

Definition of server virtualization:-

server virtualization is the virtual mechenc each machine can run it own operating system.

Role of server virtualization in cloud computing

* **Resource utilization:-** It allows multple virtual Machines to share the resources of one Physical server.
* **Cost Reduction:-** It reduces the need for physical hardware and lowers maintenances costs.
* **Scalability:-** virtual machines can be created or removed according to user requirements.

Flexibility:- Different operating systems and application can run on the same physical server

Resource Management:- CPU, memory, and storage can be allocated efficiently.

Disaster Recovery:- virtual machines can be backed up and restored when required.

Difference Between virtualization and cloud computing

Virtualization	Cloud Computing
* It is an enabing technology * It creates virtual versions of physical resources * It can operate with in a single organization Example:- VMware	* It is service delivey model * It provides computing resources over the internet. * It provides servics to wes on damand Example : Amazon web

Conclusion:- server virtualization is an important technology that improves resouse untilization and reduces costs. cloud computing uses virtualization and other technologies to deliver flexible, scalable and no-demand services

Discuss the Role of Data Centers in Delivering cloud services.

Introduction :-

A data center is a facility that contains servers, storage devices, networking equipment, and supporting infrastructed. data center are essential for storing, processing, managing and delivering cloud services.

Roles of Data Centers in cloud Computing

1. **Data Storage :-** Data centers store files, databases, application data, and backups securely.

2. **Data processing :-** servers process user requests and run cloud applications.

3. **Network Connectivity :-** networking equipment connects servers to the internet and enables users to access cloud services.

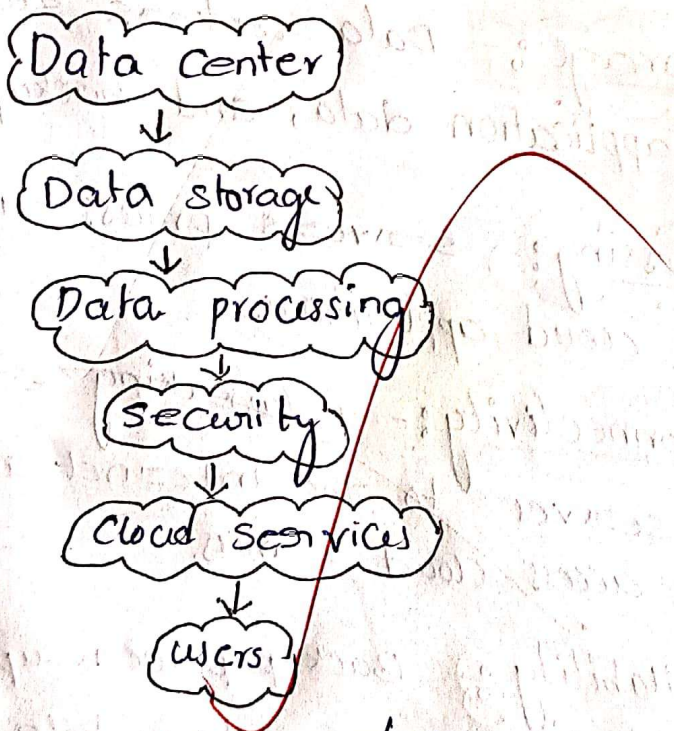
4. **high Availability :-** Backup power supplies, redundant equipment, and backup systems help reduce service interruptions.

5. **Scalability :-** Data centers provids additional computing resources when user demand increases.

Disaster Recovery :- Backup systems and geographic separate facilities help restore service after failures.

Example :- Amazon web services (AWS), microsoft Azure, and google cloud use data centers to provide service such as cloud storage, virtual servers, databases and online applications.

Role of Data centers in cloud computing



Conclusion :- Data centers form the physical foundation of cloud computing. They ensure that ensure the cloud services are available secure, scalable, and reliable for users world wide.

Differentiate Iaas, paas, and xaas with one suitable Real-coorld example for each model

Introduction :-

Cloud computing provides different service models based on the needs on the needs of users and organizations. The major models are Infrastructure as a service (Iaas), platform as a service (Paas), Software as a service (Saas) and anything as a service and ~~anything~~ as a service (xaas).

1. Infrastructure as a service (Iaas)

Paas provides virtual computing resources such as servers, storage, and networking over the Internet. Users can install and manage their operating systems and applications.

2. Platform as a service (Paas)

Paas provides a platform and development tools for building, testing, and deploying applications without managing the underlying hardware.

Ex:- Google App Engine, which provides a platform for deploying applications.

3. Software as a service (Saas)

Saas provides ready-to-use software over

users can access applications through a web browers without intalling or maintaing the underlying infrastructure.

fn:- Gmail, which allows users to send and receive emails online.

4. Anything as a service (xaas)

Xaas is a broad term that refers to delivering different types of IT resources and services over the internet on demand. It can include Infrastructue, software, platforms, storage and security services.

Comparison of cloud service models :-

Feature	Iaas	Paas	Saas	Xaas
full form	Infrastructure as a service	Platform as a service	software as a service	anything as a service
main purpose	Provides infrastru	Provides development Platforms	provides ready	provides various service

Conclusion :- Iaas provides Infrastructure, Paas Provides application development platforms, and Sas provides read-to-us software.